Retarded fields from potentials

$$\vec{E} = -\vec{\nabla}\phi - \frac{\partial \vec{A}}{\partial t}$$

$$\vec{B} = \vec{\nabla}\times\vec{A}$$

$$\varphi(\vec{r},t) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho(\vec{r}\,',t-\frac{R}{c})}{R}\,d\vec{r}\,' \qquad \vec{R} = \vec{r}-\vec{r}\,'$$

$$\vec{A}(\vec{r},t) = \frac{\mu_0}{4\pi}\int \frac{\vec{j}(\vec{r}\,',t-\frac{R}{c})}{R}\,d\vec{r}\,'$$

$$\vec{\nabla}_r\varphi = \frac{1}{4\pi\varepsilon_0}\int\left((\vec{\nabla}_r\rho)\frac{1}{R} + \rho\,\vec{\nabla}_r\frac{1}{R}\right)d\vec{r}\,'$$

$$\vec{\nabla}\frac{1}{R} = -\frac{1}{R^2}\vec{\nabla}R = -\frac{1}{R^2}\frac{\vec{R}}{R} = -\frac{\vec{R}}{R^3}$$

$$\vec{\nabla}_r\rho(\vec{r}\,',t-\tfrac{R}{c}) = \frac{\partial\rho}{\partial t}\vec{\nabla}_r(t-\tfrac{R}{c}) = \frac{\partial\rho}{\partial t}\left(-\frac{\vec{R}}{Rc}\right)$$

$$\vec{\nabla}\varphi = \frac{-1}{4\pi\varepsilon_0}\int\left(\frac{\rho(\vec{r}\,',t-\frac{R}{c})\,\vec{R}}{R^3} + \frac{\partial\rho(\vec{r}\,',t-\frac{R}{c})}{\partial t}\frac{\vec{R}}{R^2c}\right)d\vec{r}\,'$$

$$\vec{\nabla}\times\vec{A} = \frac{\mu_0}{4\pi}\int\left(\vec{\nabla}\left(\frac{1}{R}\right)\times\vec{j} + \frac{\vec{\nabla}\times\vec{j}}{R}\right)d\vec{r}\,'$$

$$\vec{\nabla}\times\vec{j}(\vec{r}\,',t-\tfrac{R}{c}) = \hat{e}_a\,\varepsilon_{abc}\frac{\partial}{\partial r_b}j_c(\vec{r}\,',t-\tfrac{R}{c})$$

$$= \hat{e}_a\,\varepsilon_{abc}\frac{\partial j_c}{\partial t}\left(-\frac{1}{c}\frac{\partial R}{\partial r_b}\right) = \hat{e}_a\,\varepsilon_{abc}\frac{R_b}{R}\frac{\partial j_c}{\partial t}\left(-\frac{1}{c}\right)$$

$$\frac{\partial R}{\partial r_b} = \frac{\partial}{\partial r_b}\sqrt{\sum_a R_aR_a} = \frac{1}{2\sqrt{\sum_a R_aR_a}}\frac{\partial}{\partial r_b}\sum_a R_aR_a = \frac{R_b}{R}$$

$$\vec{\nabla}\times\vec{A} = -\frac{\mu_0}{4\pi}\int\left(\frac{\vec{R}\times\vec{j}}{R^3} + \frac{\vec{R}\times\frac{\partial\vec{j}}{\partial t}}{cR^2}\right)d\vec{r}\,' \quad \text{where } \vec{j}(\vec{r}\,',t-\tfrac{R}{c}).$$

Continuity eqn $\frac{\partial \rho}{\partial t} + \vec{\nabla}\cdot\vec{J} = 0$

but be careful because of retardation:

$$\underset{\substack{\uparrow \\ \frac{\partial}{\partial \vec{r}'}}}{\vec{\nabla}'}\cdot\vec{J}\left(\vec{r}, t-\frac{R}{c}\right) = \underbrace{\frac{\partial \vec{J}}{\partial \vec{r}'}}_{-\frac{\partial \rho}{\partial t}} + \frac{\partial \vec{J}}{\partial t}\cdot\left(-\frac{1}{c}\underbrace{\vec{\nabla}' R}_{\frac{-\vec{R}}{R}}\right) \qquad \hat{R} = \frac{\vec{R}}{R}$$

Hence $\int \frac{\partial \rho}{\partial t}\frac{\hat{R}}{Rc}\, d\vec{r}' = -\frac{1}{c}\int d\vec{r}' \frac{(\vec{\nabla}'\cdot\vec{J})\hat{R}}{R} + \frac{1}{c^2}\int d\vec{r}' \frac{\partial \vec{J}\cdot\hat{R}}{\partial t}\hat{R}$

Consider $\frac{\vec{\nabla}'\cdot\vec{J}}{R}\hat{R} = \frac{\partial \vec{J}_j}{\partial r_j'}\frac{R_i}{R^2} = \frac{\partial}{\partial r_j'}\left(\frac{J_j R_i}{R^2}\right) -$

$$- J_j \frac{\partial}{\partial r_j'}\left(\frac{R_i}{R^2}\right) = \frac{\partial}{\partial \vec{r}_j'}\left(\frac{J_j R_i}{R^2}\right) + \frac{J_i - 2\vec{J}\cdot\hat{R}\,\hat{R}_i}{R^2}$$

$$\frac{\partial}{\partial r_j'}\frac{R_i}{R^2} = \frac{-\delta_{ij}}{R^2} - \frac{2R_i}{R^3}\underbrace{\frac{\partial R}{\partial r_j'}}_{-\frac{R_j}{R}} = \frac{-\delta_{ij} + 2\hat{R}_i\hat{R}_j}{R^2}$$

$$\Rightarrow \int d\vec{r}' \frac{\partial}{\partial \vec{r}_j}\left(\frac{J_j R_i}{R^2}\right) = 0 \text{ (surface integral)}$$

$$-\frac{1}{c}\int d\vec{r}' \frac{\vec{\nabla}'\cdot\vec{J}}{R}\hat{R} = \frac{1}{c}\int d\vec{r}' \frac{2(\vec{J}\cdot\hat{R})\hat{R} - \vec{J}}{R^2} =$$

$$= \frac{1}{c}\int d\vec{r}' \frac{(\vec{J}\cdot\hat{R})\hat{R} + (\vec{J}\times\hat{R})\times\hat{R}}{R^2}$$

$A\times(B\times C)$
$= B(A\cdot C)$
$- C(A\cdot B)$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int d\vec{r}\,'\left\{\frac{\rho\,\hat{R}}{R^2} + \frac{(\vec{J}\cdot\hat{R})\hat{R} + (\vec{J}\times\hat{R})\times\hat{R}}{cR^2}\right.$$

$$\left. + \frac{\left(\frac{\partial\vec{J}}{\partial t}\cdot\hat{R}\right)\hat{R}}{c^2 R}\right\} - \frac{\mu_0}{4\pi}\int d\vec{r}\,'\,\frac{\frac{\partial\vec{J}}{\partial t}}{R}$$

$$\mu_0 = \frac{1}{\varepsilon_0 c^2}$$

But $\left(\frac{\partial\vec{J}}{\partial t}\times\hat{R}\right)\times\hat{R} = -\left(\hat{R}\times\frac{\partial\vec{J}}{\partial t}\right)\times\hat{R} =$

$$= (-)\left\{\frac{\partial\vec{J}}{\partial t} - \hat{R}\left(\hat{R}\cdot\frac{\partial\vec{J}}{\partial t}\right)\right\}$$

$$\Rightarrow \vec{E}(\vec{r},t) = \frac{1}{4\pi\varepsilon_0}\int d\vec{r}\,'\left\{\frac{\rho\,\hat{R}}{R^2} + \frac{(\vec{J}\cdot\hat{R})\hat{R} + (\vec{J}\times\hat{R})\times\hat{R}}{cR^2}\right.$$

$$\left. + \frac{\left(\frac{\partial\vec{J}}{\partial t}\times\hat{R}\right)\times\hat{R}}{c^2 R}\right\}$$

$$\vec{B}(\vec{r},t) = +\frac{\mu_0}{4\pi}\int d\vec{r}\,'\left\{\frac{\vec{J}\times\hat{R}}{R^2} + \frac{\frac{\partial\vec{J}}{\partial t}\times\hat{R}}{cR}\right\}$$

(These are Jefimenko equations)

Most important features: (1) Near fields $\sim \frac{1}{R^2}$

(2) Far fields $\sim \frac{1}{R}$ $\Rightarrow$ this is radiation.

(3) In the far region only <u>transverse</u> components of the current contribute to <u>radiation</u> fields $\vec{E}$ and $\vec{B}$.

Far-fields $\left(\sim \frac{1}{R}\right)$

$$\vec{E}(\vec{r},t) = \frac{-1}{4\pi\varepsilon_0}\int d\vec{r}\,' \; \frac{\frac{\partial}{\partial t}\left[\vec{J}(\vec{r}\,',t-\frac{R}{c}) - \left(\vec{J}(\vec{r}\,',t-\frac{R}{c})\cdot\hat{R}\right)\hat{R}\right]}{c^2 R}$$

$$= -\frac{\mu_0}{4\pi}\int d\vec{r}\,' \; \frac{\left(\frac{\partial}{\partial t}\vec{J}(\vec{r}\,',t-\frac{R}{c})\times\hat{R}\right)\times\hat{R}}{R}$$

$$\vec{B}(\vec{r},t) = \frac{\mu_0}{4\pi}\int d\vec{r}\,' \; \frac{\frac{\partial}{\partial t}\vec{J}(\vec{r}\,',t-\frac{R}{c})\times\hat{R}}{cR} =$$

$$= \frac{\mu_0}{4\pi c}\int d\vec{r}\,' \; \frac{\frac{\partial}{\partial t}\left[\vec{J}(\vec{r}\,',t-\frac{R}{c}) - \left(\vec{J}(\vec{r}\,',t-\frac{R}{c})\cdot\hat{R}\right)\hat{R}\right]\times\hat{R}}{R}$$

Both fields are expressed in terms of $\frac{\partial}{\partial t}\vec{J}_\perp(\vec{r}\,',t-\frac{R}{c})$

where $\vec{J}_\perp \equiv \vec{J} - (\vec{J}\cdot\hat{R})\hat{R} = -(\vec{J}\times\hat{R})\times\hat{R} = (\hat{R}\times\vec{J})\times\hat{R}$

is transverse current.

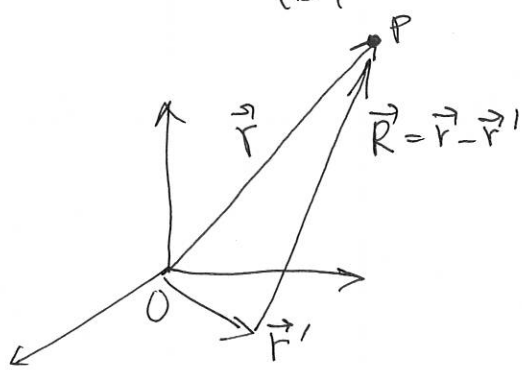
$\vec{J}_\perp$ is vector orthogonal to $\hat{R}$.

Hence we see: (a) $\vec{E}\perp\hat{R}$, $\hat{B}\perp\hat{R}$ and $\vec{E}\perp\vec{B}$

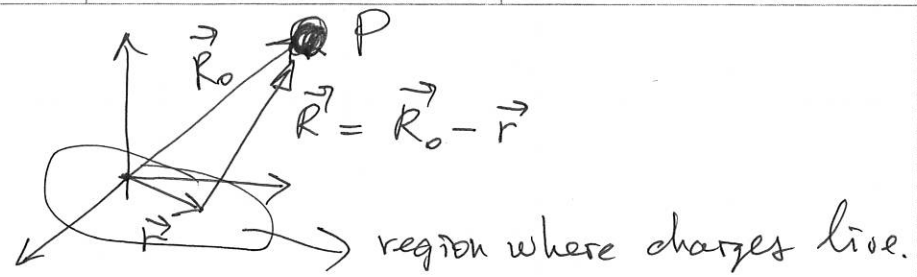
(plane wave structure)

(b) $\frac{|\vec{E}|}{|\vec{B}|} = c$. (for $\vec{R} = \vec{r}-\vec{r}\,' \approx \vec{r}$)

$r, R \gg r'$



let now



For $R_0 \gg r$ (far field condition) $\vec{R} \approx \vec{R}_0$

$\Rightarrow \hat{R} = \hat{R}_0$ in which case (let $\hat{n} = \hat{R}_0 = \frac{\vec{R}_0}{R_0}$)

$$\vec{B} \cong \frac{\mu_0}{4\pi} \frac{1}{c} \int d\vec{r}\,' \frac{1}{R_0} \frac{\partial \vec{J}(\vec{r}\,', t - \frac{R}{c})}{\partial t} \times \hat{n} =$$

$$= \frac{1}{c} \frac{\partial}{\partial t} \vec{A}(\vec{r}, t) \times \hat{n}$$

$$\vec{E}(r,t) \approx \left\{ \left(\frac{1}{4\pi\varepsilon_0} \int d\vec{r}\,' \frac{\frac{\partial \vec{J}}{\partial t}}{c^2 R_0} \right) \times \hat{n} \right\} \times \hat{n}$$

$$\left(\frac{1}{\varepsilon_0 c^2} = \mu_0 \right) \quad = \left\{ \frac{\partial}{\partial t} \vec{A}(r,t) \times \hat{n} \right\} \times \hat{n}$$

so that $\vec{B} = \frac{1}{c} \hat{n} \times \vec{E}$